

The IMDb logo, featuring the letters "IMDb" in a bold, black, sans-serif font. The letters are set against a solid yellow rectangular background. The "i" is lowercase and has a distinctive dot.

IMDb

**IS THERE ANY
CORRELATION
BETWEEN
GROSS
REVENUE AND
RATING?**

PROJECT BACKGROUND

The IMDb logo is displayed in a large, bold, black font on a solid yellow rectangular background. The letters are thick and stylized, with the 'b' having a rounded, lowercase-like appearance.

This is an exploratory data analysis of the top 20 movies spanning 25 years (1989-2014) to see if there is any correlation between gross revenue and ratings according to critics. IMDb determines the “top 20” as movies that have high grossing revenues and/or critical acclaim.

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CATEGORICAL TOTALS DASHBOARD

CATEGORICAL TOTALS

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- ⌘ Dashboard summary of budget, gross revenue, ratings, and runtime
- ⌘ Top 10 movies for each category
- ⌘ Number of movies categorized by genre
- ⌘ Number of movies categorized by MPAA rating

CATEGORICAL TOTALS DASHBOARD

SUM OF BUDGETS

\$42.99bn

SUM OF GROSS

\$195.43bn

SUM OF RUNTIME (MIN)

60K

TOP MOVIE RATING

9.00

TOP 10 MOVIE BUDGETS

Pirates of ...	\$380.00M
Pirates of ...	\$300.00M
Godzilla	\$290.00M
Superma...	\$270.00M
Tangled	\$260.00M
Spider-M...	\$258.00M
Harry Pot...	\$250.00M
Harry Pot...	\$250.00M
The Dark ...	\$250.00M
The Hobb...	\$250.00M
The Hobb...	\$250.00M

TOP 10 GROSSING MOVIES

Avatar	\$2.8bn
Titanic	\$1.8bn
The Aven...	\$1.5bn
Harry Pot...	\$1.3bn
Frozen	\$1.3bn
Iron Man 3	\$1.2bn
Transform...	\$1.1bn
The Lord ...	\$1.1bn
Skyfall	\$1.1bn
The Dark ...	\$1.1bn

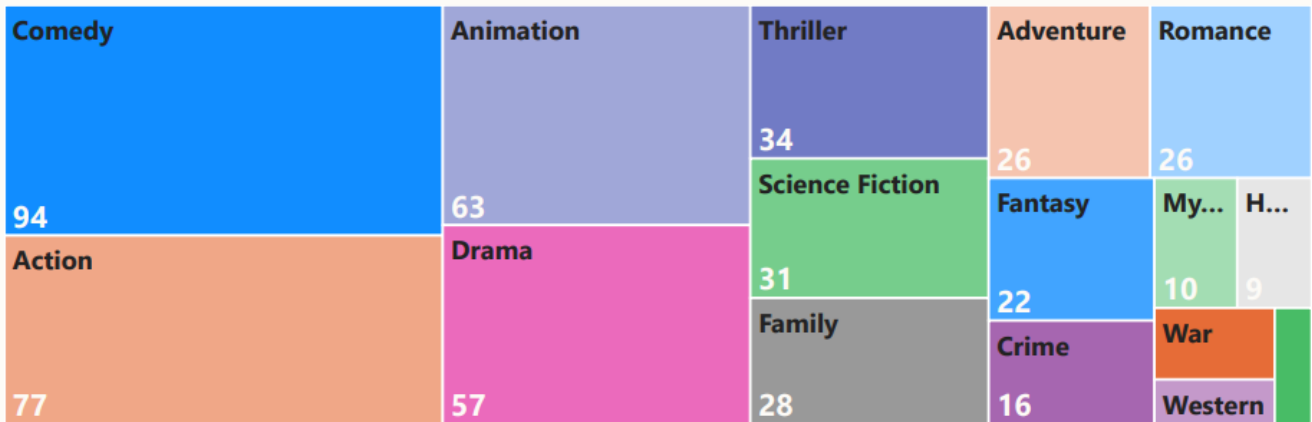
TOP 10 MOVIE RUNTIMES

Godzilla	262
The Lord ...	201
Schindler...	195
Teenage ...	194
Titanic	194
JFK	189
The Gree...	189
King Kong	187
Jurassic P...	184
Pearl Har...	183

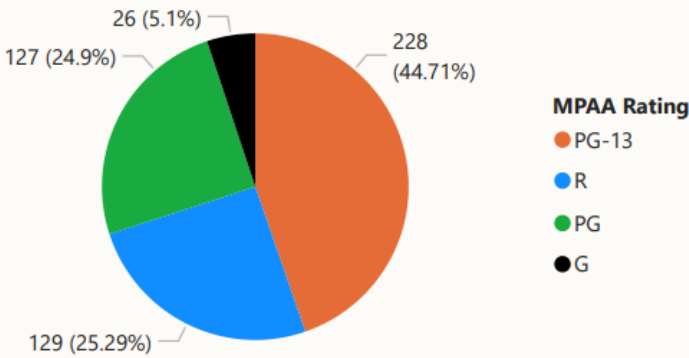
TOP 10 RATED MOVIES

The Dark ...	9.0
Pulp Ficti...	8.9
Schindler...	8.9
The Lord ...	8.9
Fast and ...	8.8
Forrest G...	8.8
Inception	8.8
The Lord ...	8.8
The Lord ...	8.7
The Matrix	8.7

COUNT BY GENRE



COUNT BY MPAA RATING



CATEGORICAL TOTALS SUMMARY

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- ⌘ Sum of all budgets: \$42.99B
- ⌘ Sum of gross revenue: \$195.43B
- ⌘ Sum of runtime: 60k minutes
- ⌘ Highest rating: Dark Night with 9.0
- ⌘ Biggest budget: Pirates of the Caribbean: On Stranger Tides at \$380M
- ⌘ Most gross revenue: Avatar at \$2.8B
- ⌘ Longest run time: Godzilla at 262 minutes
- ⌘ Most popular genre: Comedy
- ⌘ Most MPAA ratings: PG-13

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**CORRELATIONS
DASHBOARD**

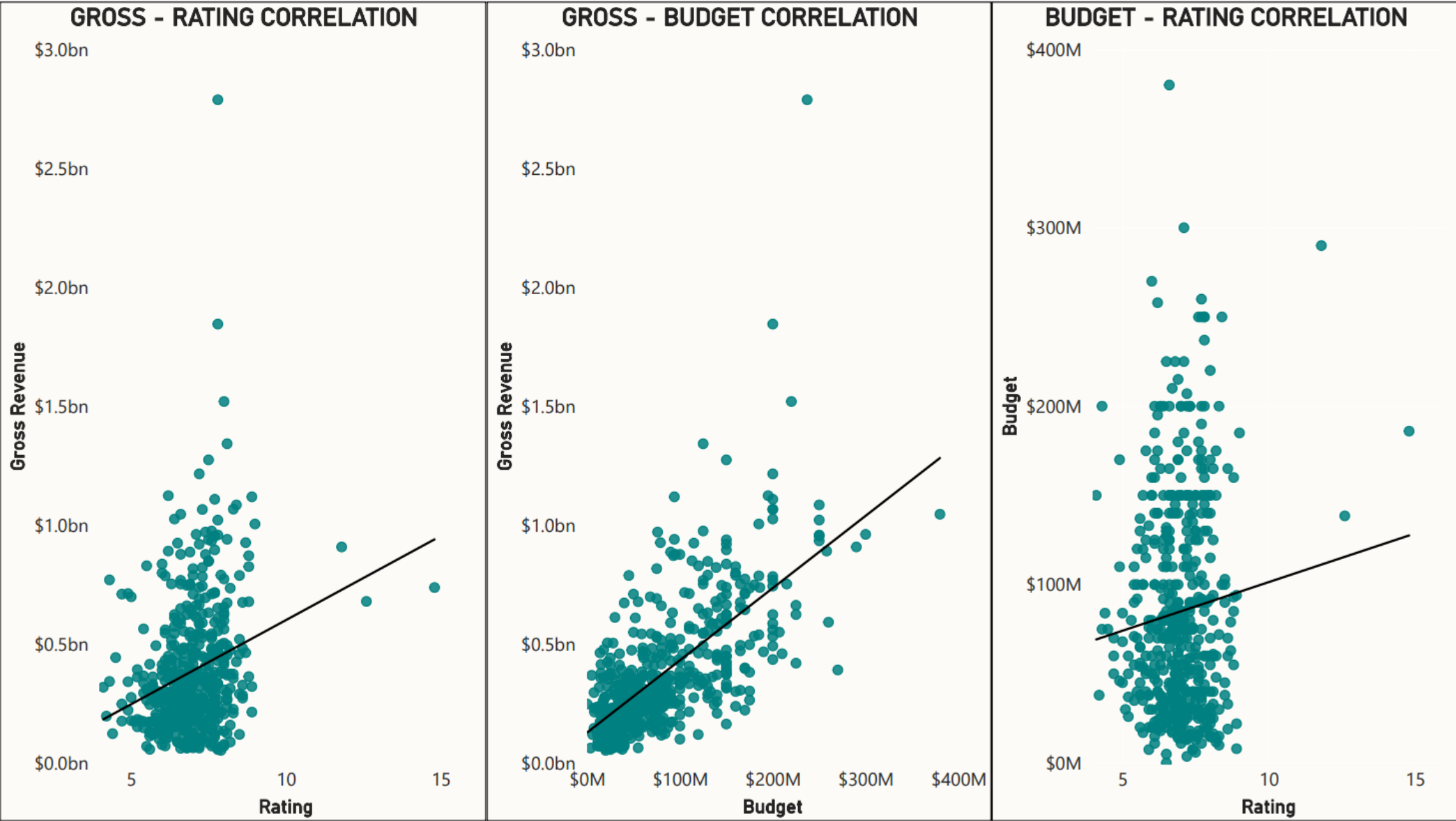
CORRELATIONS DASHBOARD

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A dashboard showing potential correlations between the following features:

- ⌘ Gross Revenue and Rating
- ⌘ Gross Revenue and Budget
- ⌘ Budget and Rating

CORRELATIONS DASHBOARD



CORRELATIONS SUMMARY

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- ⌘ Gross and Rating: slight positive correlation indicating the higher the rating, the more revenue a movie generates.
- ⌘ Gross and Budget: slight positive correlation indicating the higher the budget, the higher the revenue.
- ⌘ Budget and Rating: Very low positive correlation indicating a very slight probability of a higher budget equating to a better rating.

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**IN-DEPTH
ANALYSIS
DASHBOARD**

IN-DEPTH ANALYSIS DASHBOARD

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- ⌘ This dashboard features minimum, maximum, average, median data for budget, gross revenue, and rating
- ⌘ Lists budget, gross, and rating data for each movie sorted in ascending order according to the numeric value of the category
- ⌘ Multiple slicers of each category provide in-depth insights especially when combined

IN-DEPTH ANALYSIS DASHBOARD

\$60,000.00

Minimum Budget 🟢

\$53,000,000.00

Minimum Gross 💰

4.10

Minimum Rating ⭐

\$84,299,050.07

Average Budget 🟢

\$383,198,716.58

Average Gross 💰

6.92

Average Rating ⭐

\$380,000,000.00

Maximum Budget 🟢

\$2,787,965,087.00

Maximum Gross 💰

9.00

Maximum Rating ⭐

\$70,500,000.00

Median Budget 🟢

\$310,071,633.00

Median Gross 💰

6.90

Median Rating ⭐

BUDGET BY MOVIE

Pirates of the ...	\$380.00M
Pirates of the ...	\$300.00M
Godzilla	\$290.00M
Superman Ret...	\$270.00M
Tangled	\$260.00M
Spider-Man 3	\$258.00M
Harry Potter a...	\$250.00M
Harry Potter a...	\$250.00M
The Dark Knig...	\$250.00M
The Hobbit: A...	\$250.00M
The Hobbit: Th...	\$250.00M
Avatar	\$237.00M
Man of Steel	\$225.00M
Men in Black 3	\$225.00M
The Chronicles...	\$225.00M
The Avengers	\$220.00M
The Amazing ...	\$215.00M
X-Men: The La...	\$210.00M
King Kong	\$207.00M

GROSS BY MOVIE

Avatar	\$2,787.97M
Titanic	\$1,845.03M
The Avengers	\$1,519.56M
Harry Potter a...	\$1,342.00M
Frozen	\$1,274.22M
Iron Man 3	\$1,215.44M
Transformers: ...	\$1,123.75M
The Lord of th...	\$1,118.89M
Skyfall	\$1,108.56M
The Dark Knig...	\$1,084.94M
Toy Story 3	\$1,066.97M
Pirates of the ...	\$1,065.66M
Pirates of the ...	\$1,045.71M
Alice in Wonde...	\$1,025.49M
The Hobbit: An...	\$1,021.10M
The Dark Knight	\$1,004.56M
Harry Potter a...	\$975.10M
Despicable Me 2	\$970.76M
Pirates of the ...	\$961.00M

RATING BY MOVIE

The Dark Kni...	9.0
Pulp Fiction	8.9
Schindler's List	8.9
The Lord of t...	8.9
Fast and Furi...	8.8
Forrest Gump	8.8
Inception	8.8
The Lord of t...	8.8
The Lord of t...	8.7
The Matrix	8.7
Interstellar	8.6
Saving Privat...	8.6
Se7en	8.6
The Green M...	8.6
The Silence ...	8.6
Gladiator	8.5
Star Trek: Ge...	8.5
Terminator 2:...	8.5
The Departed	8.5

Genre, Title

☐ Select all

✓ ☐ Action

✓ ☐ Adventure

✓ ☐ Animation

✓ ☐ Comedy

Runtime

79

201



MPAA Rating

☐ Select all

☐ G

☐ PG

☐ PG-13

☐ R

Rating

4.10

9.00



Year, Quarter, Month, Day

☐ Select all

✓ ☐ 1989

✓ ☐ 1990

✓ ☐ 1991

✓ ☐ 1992

Rating Count

974

2127228



IN-DEPTH ANALYSIS SUMMARY

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- ⌘ Budget: Min \$60,000, Max \$380,000,000, Avg \$84,299,050, Median \$70,500,000, Range \$379,940,000
- ⌘ Gross: Min \$53,000, Max \$2,787,965,087, Avg \$383,198,716, Median \$310,071,633, Range \$2,787,912,087
- ⌘ Rating: Min 4.10, Max 9.0, Avg 6.92, Median 6.9, Range 4.9

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**BUDGET VS
GROSS BY
YEAR**

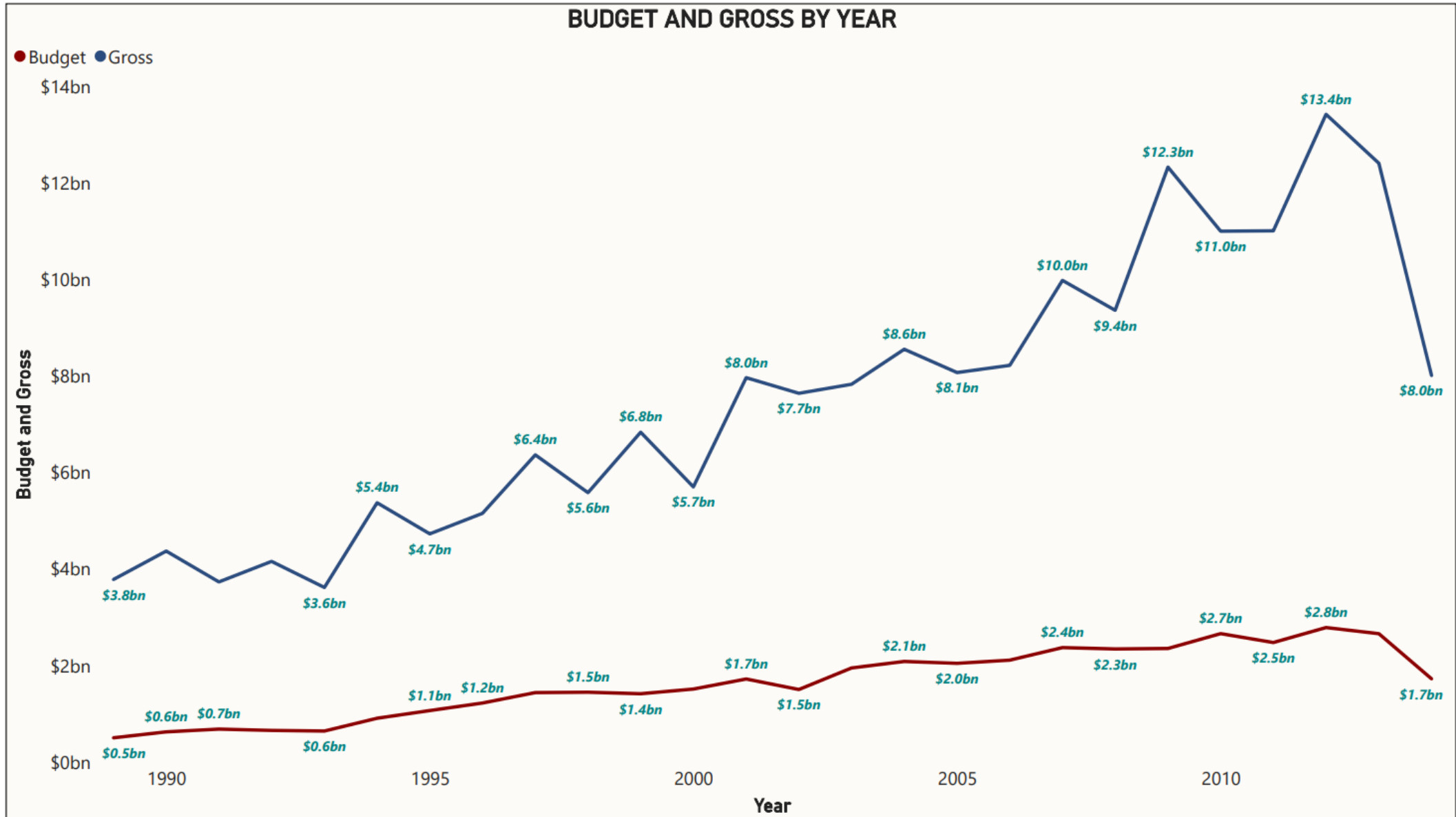
BUDGET VS GROSS BY YEAR

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Dual-line chart where the budgets and gross revenue are assessed over the years

BUDGET VS GROSS BY YEAR



BUDGET VS GROSS SUMMARY

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- 🎬 2012 was the year with the biggest combined budget at \$2.8B
- 🎬 2012 was also the year with the most combined gross revenue at \$13.4B

IMDb

CONCLUSION

SUMMARY

“Is there any correlation between gross revenue and rating?” Speaking in a general sense, I believe so. Aside from the “Correlation” visualization, here are some of the insights that helped me with my conclusion:

- ⊗ According to the data, the average gross revenue is \$383.20M and the average rating is 6.92 (and the median is 6.90) on a scale of 1-10. This tells me that the average film rating is above average, almost a 7, for movies that grossed an average amount of \$383.20M (a substantial amount).
- ⊗ The minimum rating was 4.10 and the maximum rating was a 9.0. This tells me that even though the lowest rating was below average, it wasn't extremely below average, it was just below a 5.
- ⊗ The highest grossing movie was “Avatar,” grossing \$2.79B, and rated at a 7.8. This tells me that the highest grossing movie has a relatively high rating.
- ⊗ The lowest movie rating belonged to “The Last Airbender,” rated at 4.10. This is below the average rating of 6.92. This movie grossed \$318,502,923.00. This is below the average of \$383.20M. Even though this movie grossed a significant amount of money, it was still below average, and it correlates with its below average rating.

FURTHER ANALYSIS

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- ⌘ A dataset that covers more years and movies with all applicable data
- ⌘ Using the slicers implemented in the “In-Depth Analysis” dashboard to explore all possible combinations and outcomes
- ⌘ Search for trends according to the days of the week